

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Medium

Question

Average Nitrate and Phosphate Concentrations in Seawater after Volcanic Eruption

Nutrient	Seawater in lava-affected area, 5–45 meters below surface	Seawater in lava-affected area, 75–125 meters below surface	Seawater outside of lava-affected area, 5–45 meters below surface	Seawater outside of lava-affected area, 75–125 meters below surface
Nitrate (micromoles per liter)	3.1	0.4	≤0.03	≤0.01
Phosphate (micromoles per liter)	0.17	0.09	0.14	0.06

After a volcanic eruption spilled lava into North Pacific Ocean waters, a dramatic increase of diatoms (a kind of phytoplankton) near the surface occurred. Scientists assumed the diatoms were thriving on nutrients such as phosphate from the lava, but analysis showed these nutrients weren't present near the surface in forms diatoms can consume. However, there was an abundance of usable nitrate, a nutrient usually found in much deeper water and almost never found in lava. Microbial oceanographer Sonya Dyhrman and colleagues believe that as the lava plunged nearly 300 meters below the surface it dislodged pockets of this nutrient, releasing it to float upward, given that _____

Which choice most effectively uses data from the table to complete the statement?

Answer

- A. at 5–45 meters below the surface, the average concentration of phosphate was about the same in the seawater in the lava-affected area as in the seawater outside of the lava-affected area.
- B. for both depth ranges measured, the average concentrations of nitrate were substantially higher in the seawater in the lava-affected area than in the seawater outside of the lava-affected area.
- C. for both depth ranges measured in the seawater in the lava-affected area, the average concentrations of nitrate were substantially higher than the average concentrations of phosphate.
- D. in the seawater outside of the lava-affected area, there was little change in the average concentration of nitrate from 75–125 meters below the surface to 5–45 meters below the surface.

Correct Answer: B

Rationale

Choice B is the best answer. The claim is that the lava freed the deep nitrate, allowing it to float upward. The table supports this by showing that there was more nitrate in the lava-affected seawater at various depths above 300 meters (the depth to which the lava plunged) than in unaffected seawater.

Choice A is incorrect. This choice doesn't complete the statement in a way that supports the claim. The claim is about nitrate, not phosphate. Choice C is incorrect. This choice doesn't complete the statement in a way that supports the claim. The claim is only about nitrate—the concentrations of phosphate aren't relevant. Choice D is incorrect. This choice doesn't complete the statement in a way that supports the claim. It doesn't say anything about the seawater inside the lava-affected area.